

IngeniumAI Hackathon – Non-contrast aorta & PA segmentation
Compiled 2026-08-14 11:01 UTC

Task: segment pulmonary artery and aorta on non-contrast chest CT
so PHAST can run earlier than contrast CTPA.

Method: contrast-removal domain adaptation (not plain supervised).
Source = unused contrast CTPA. Vessel HU is flattened to soft-tissue
range so SegResNet must use shape and location. Mixed with held-in
non-contrast cases under the same HU randomization.
Val = 5 held-out non-contrast cases (never trained on).

Device: mps
Non-contrast paired: 5 Contrast paired: 0
Epochs: 0 steps/epoch: 32
PA-focused sampling: pos=0.9 PA frac=0.7
Class weights (bg, PA, aorta): [0.4, 2.5, 1.0]

Mean hold-out Dice (non-contrast val)
Baseline PA 0.000 aorta 0.045 mean 0.023
Adapted PA 0.703 aorta 0.725 mean 0.714

Clinical mean equivalent diameter at max-PA GT slice
GT MPA 61.0 mm aorta 39.6 mm ratio 1.6358178359962985
Base MPA 0.0 mm aorta 8.6 mm ratio None
Adapt MPA 55.7 mm aorta 40.0 mm ratio 1.4056915672340522

Licence: CT-RATE is CC BY-NC-SA 4.0 (demonstration only).
A production PHAST model would be retrained on licensed/clinical data.
FUMPE / CAD-PE contrast sets are CC BY 4.0.

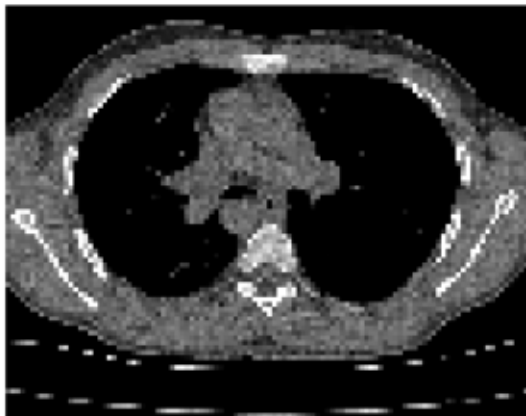
testset baseline train 10035 a 1.nii

testset_baseline train_10035_a_1.nii PA Dice 0.00 aorta 0.00

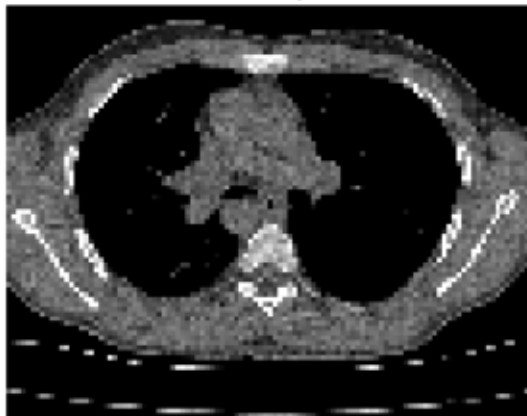
Ground truth



Prediction



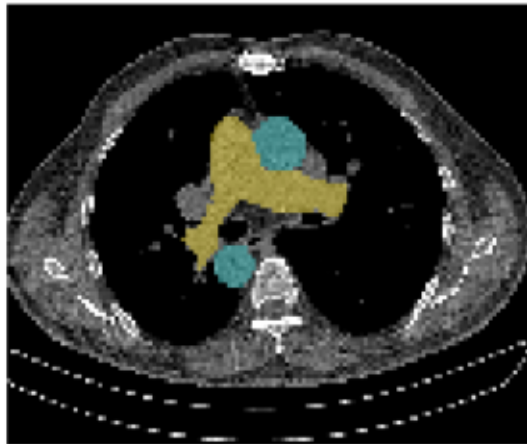
CT only



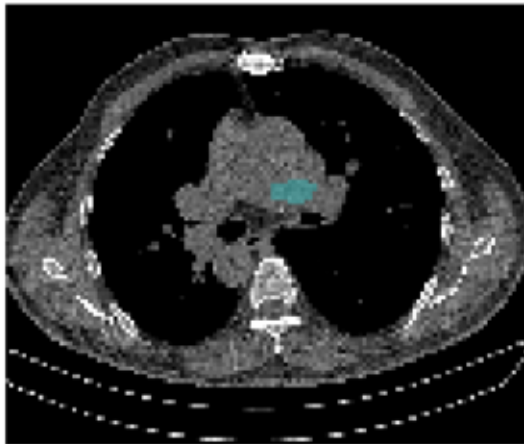
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testset_baseline train_10036_a_2.nii PA Dice 0.00 aorta 0.00

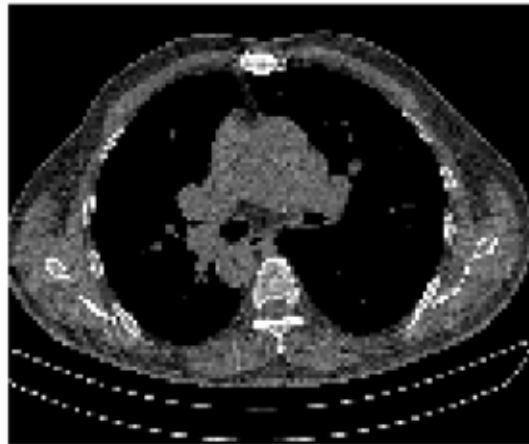
Ground truth



Prediction



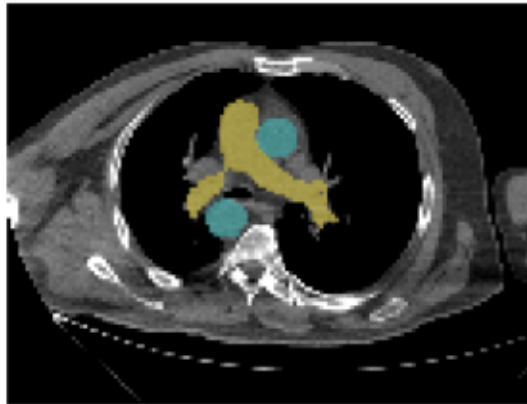
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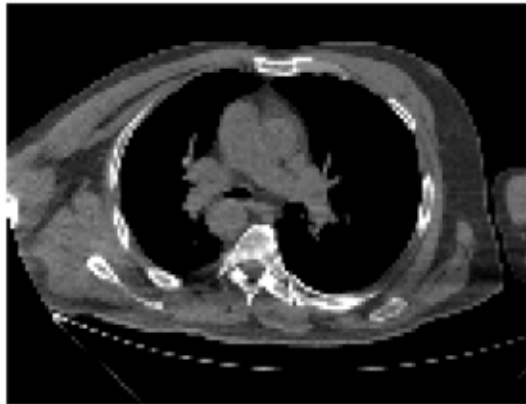
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testset_baseline train_10037_a_2.nii PA Dice 0.00 aorta 0.00

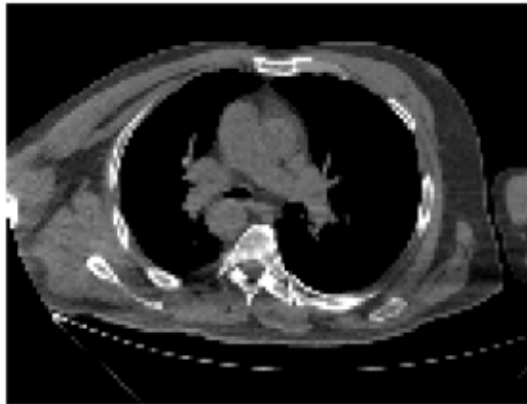
Ground truth



Prediction



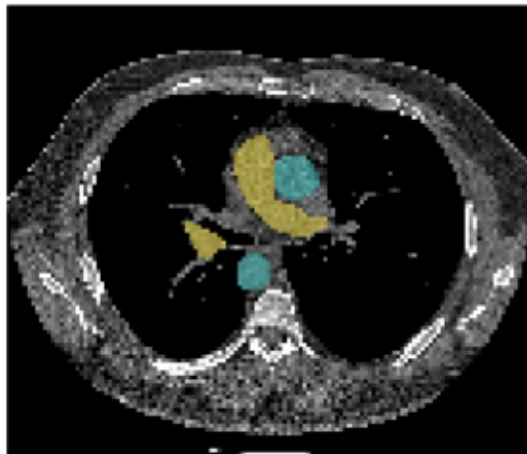
CT only



testset baseline train 10038 a 2.nii

testset_baseline train_10038_a_2.nii PA Dice 0.00 aorta 0.23

Ground truth



Prediction



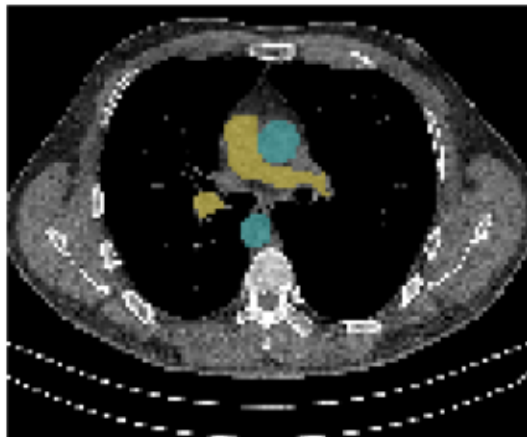
CT only



testset baseline train 10039 a 2.nii

testset_baseline train_10039_a_2.nii PA Dice 0.00 aorta 0.00

Ground truth



Prediction



CT only



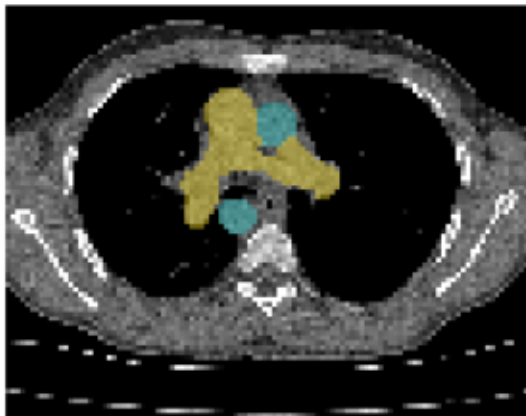
testset adapted train 10035 a 1.nii

testset_adapted train_10035_a_1.nii PA Dice 0.65 aorta 0.60

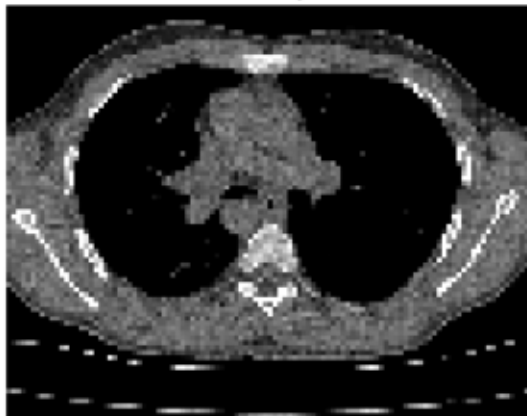
Ground truth



Prediction



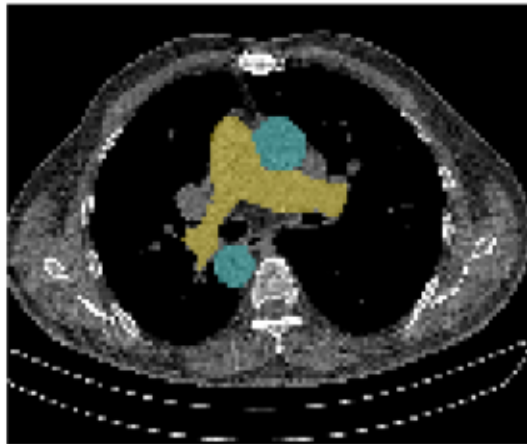
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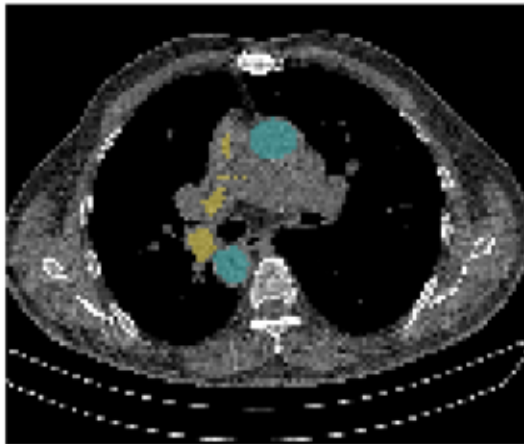
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testset_adapted train_10036_a_2.nii PA Dice 0.39 aorta 0.70

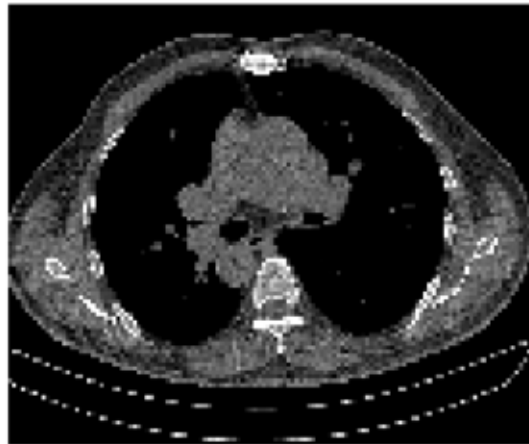
Ground truth



Prediction



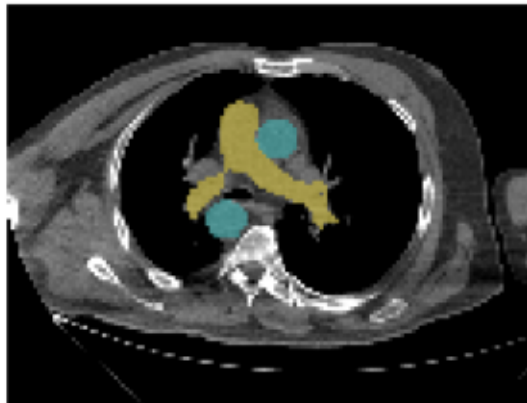
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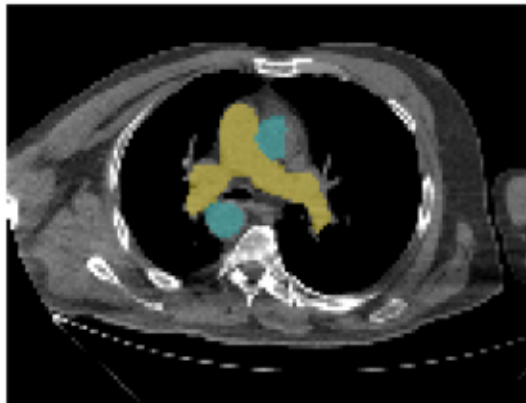
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testset_adapted train_10037_a_2.nii PA Dice 0.85 aorta 0.73

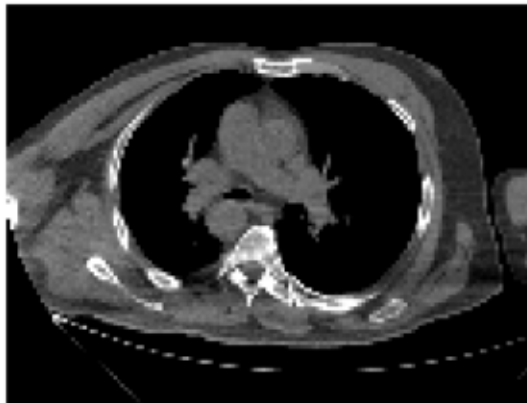
Ground truth



Prediction



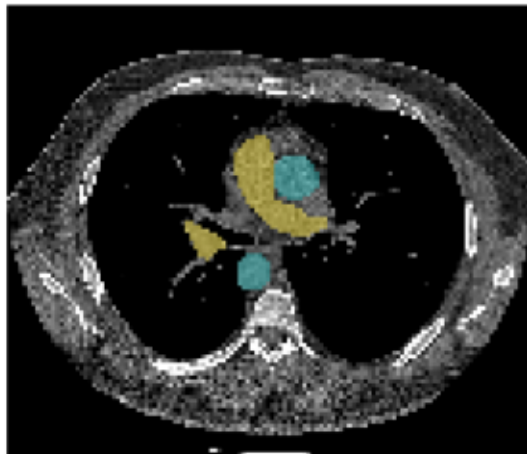
CT only



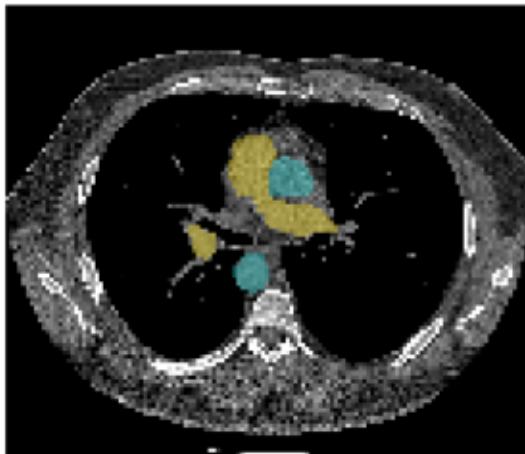
testset adapted train 10038 a 2.nii

testset_adapted_train_10038_a_2.nii PA Dice 0.80 aorta 0.79

Ground truth



Prediction



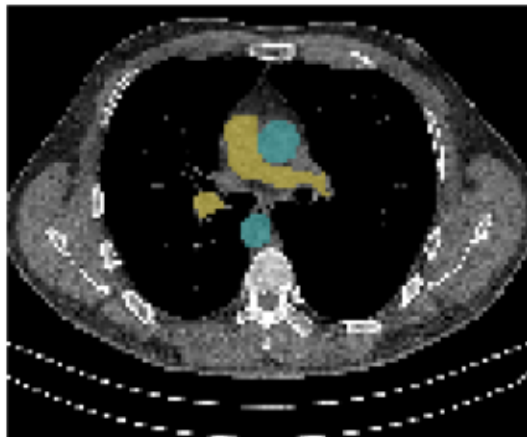
CT only



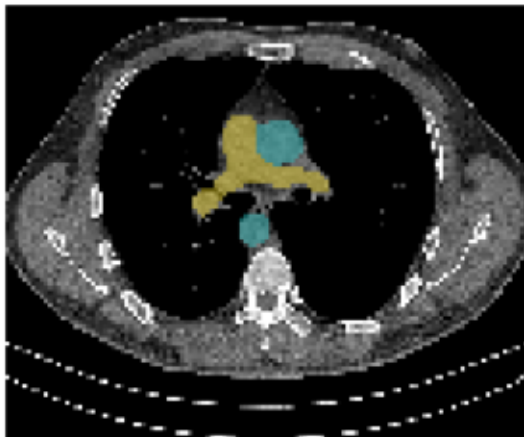
testset adapted train 10039 a 2.nii

testset_adapted train_10039_a_2.nii PA Dice 0.83 aorta 0.81

Ground truth



Prediction



CT only

